adapted on
Lynford L. Goddard
Brian T. Cunningham

Lectures 6-7

Boundary Conditions

Poisson's and Laplace's Equation

Victoria Shao
yangshao@Illinois.edu

© Lynford Goddard and Brian Cunningham - Copying, publishing or distributing without express written permission is prohibited.

Laws of Electrostatics

$$\nabla \cdot \vec{\boldsymbol{D}} = \rho$$

$$\nabla \times \vec{\boldsymbol{E}} = 0$$

$$\vec{\boldsymbol{D}} = \epsilon_0 \vec{\boldsymbol{E}}$$

Or

$$\nabla \cdot \epsilon_0 \vec{\boldsymbol{E}} = \rho$$

$$\nabla \times \vec{\boldsymbol{E}} = 0$$

$\vec{\boldsymbol{D}}$ and $\vec{\boldsymbol{E}}$ will in general be continuous functions of position $\vec{\boldsymbol{r}}$ except at boundary surfaces (contain surface charge density $\rho_s(\vec{\boldsymbol{r}})$)

Maxwell's Eqns

only within one material

Faraday's Law $\oint_C \vec{E} \bullet d\vec{l} = -\frac{d}{dt}\iint_S \vec{B} \bullet d\vec{S}$ $\quad \nabla \times \vec{E} = -\frac{d\vec{B}}{dt}$

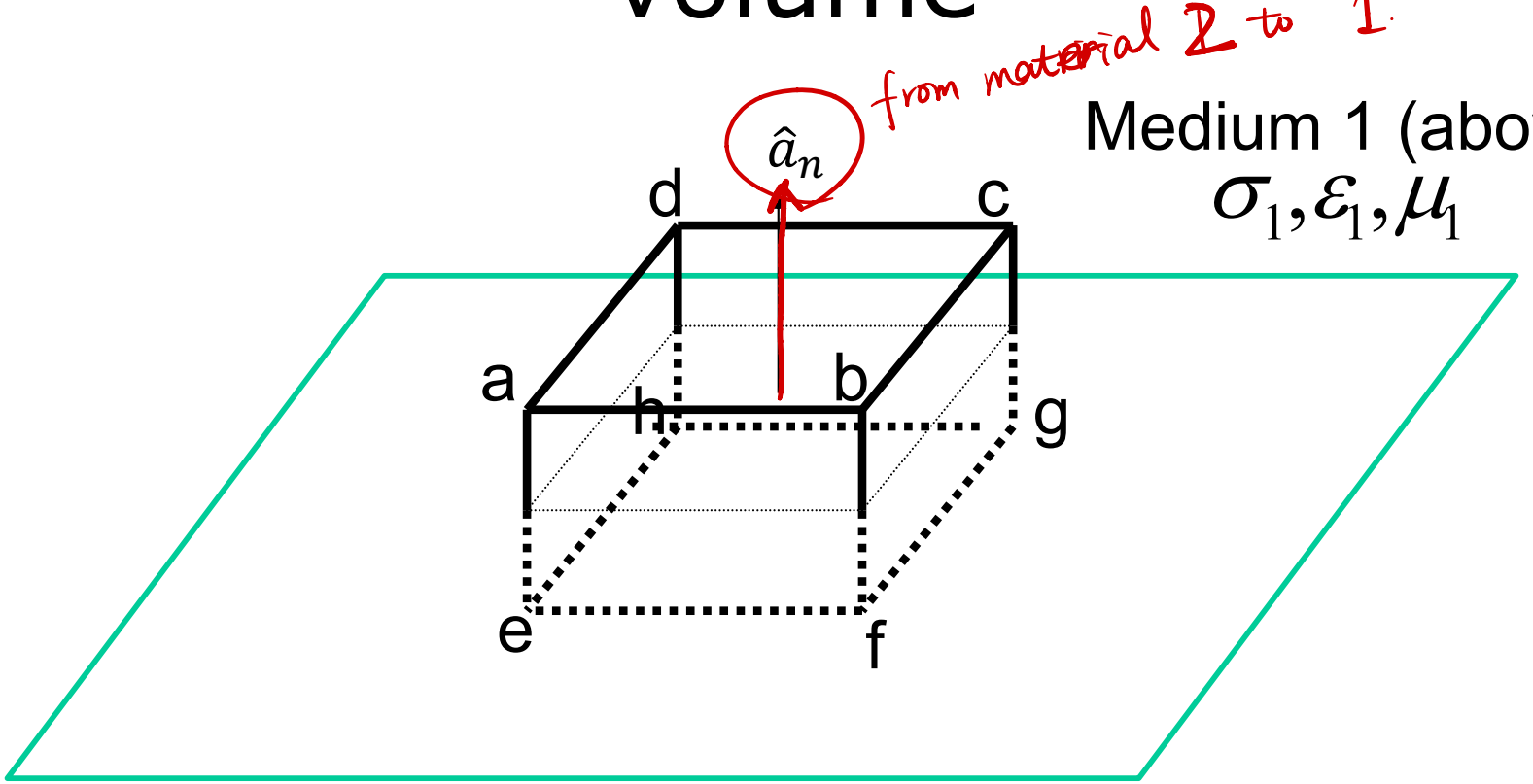
Ampere's Law $\oint_C \vec{H} \bullet d\vec{l} = \iint_S \vec{J} \bullet d\vec{S} + \frac{d}{dt}\iint_S \vec{D} \bullet d\vec{S}$ $\quad \nabla \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}$

Gauss' Law $\oiint_S \vec{B} \bullet d\vec{S} = 0$ $\quad \nabla \bullet \vec{B} = 0$

Gauss' Law $\oiint_S \vec{D} \bullet d\vec{S} = \iiint_V \rho dV$ $\quad \nabla \bullet \vec{D} = \rho$

Integral forms are valid for ALL closed paths and closed surfaces, EVEN WHEN THEY SPAN A BOUNDARY BETWEEN TWO MATERIALS

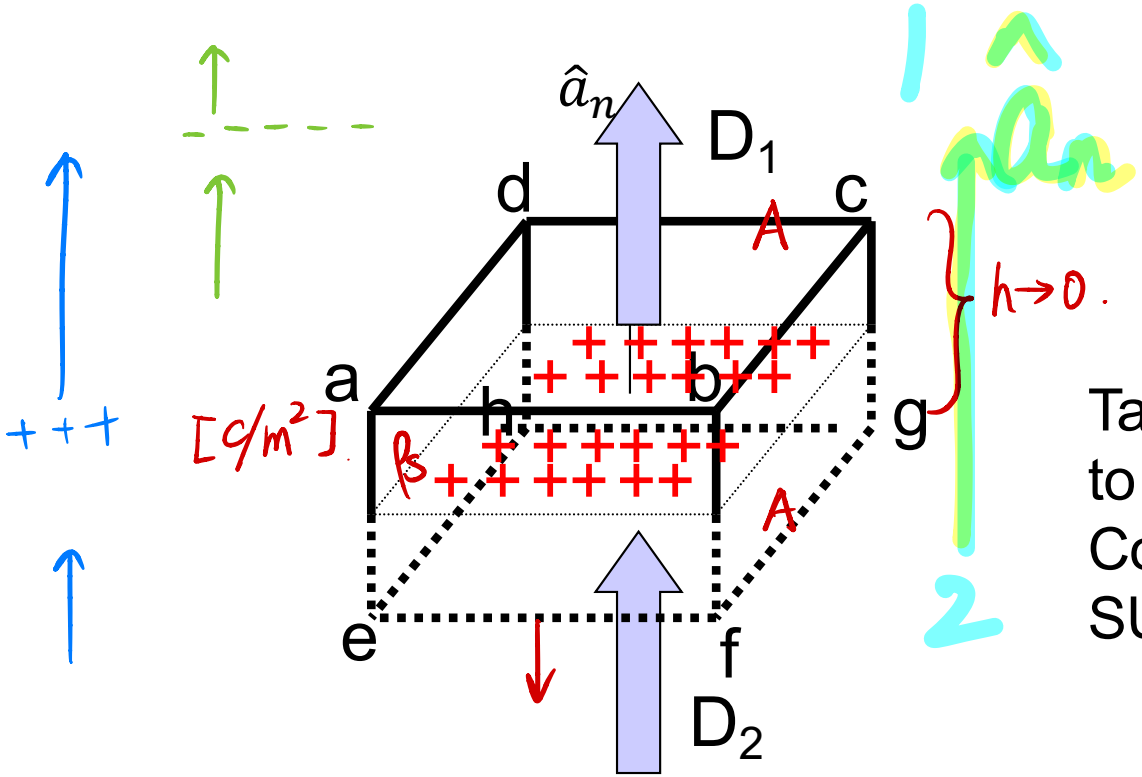
Closed surface through the volume



Closed surface: 6 sides of the cube

Apply Gauss's Law to the closed surface

Gauss' Law for D at Boundary



$$\oiint_S \vec{D} \bullet d\vec{S} = \iiint_V \rho dV$$

Top $D_1 A$ Bottom $-D_2 A = \rho_s \cdot A$.

Take limit as *ae*, *bf*, *cg*, and *dh* go to zero

Consider D_1 and D_2 NORMAL TO SURFACE

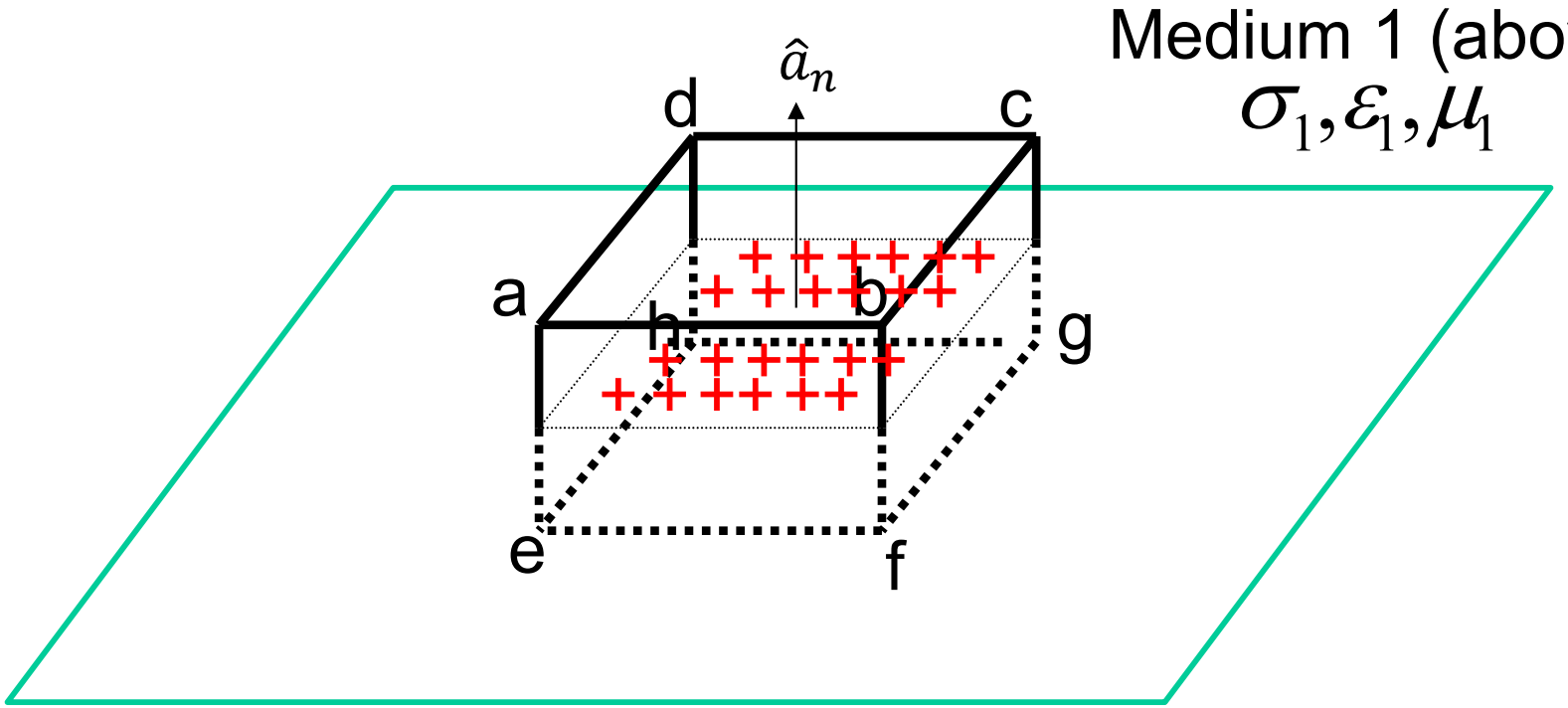
$$D_1 - D_2 = \rho \text{ i.e. } D_n \text{ is discontinuous because of } \rho_S$$

$$D_{n1} - D_{n2} = \rho_s$$

$$\hat{a}_n \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s$$

Note: we know nothing about D_t

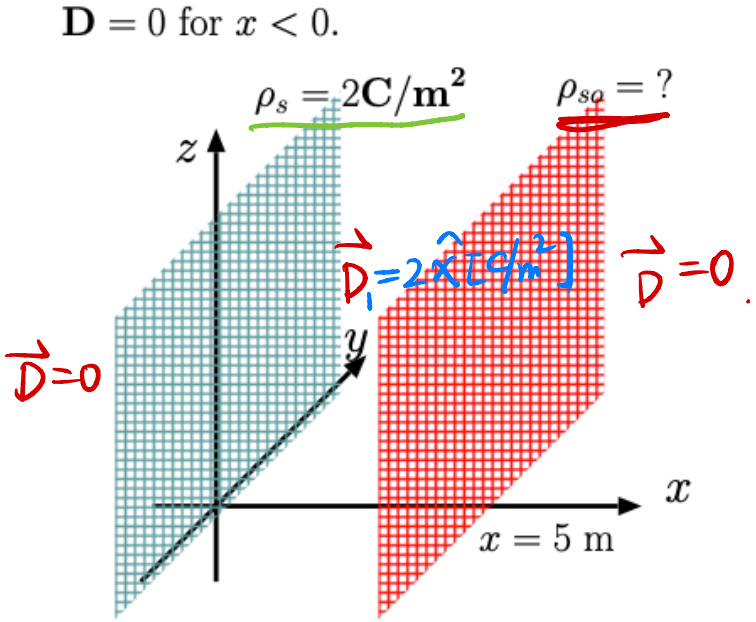
The closed volume can enclose surface charges

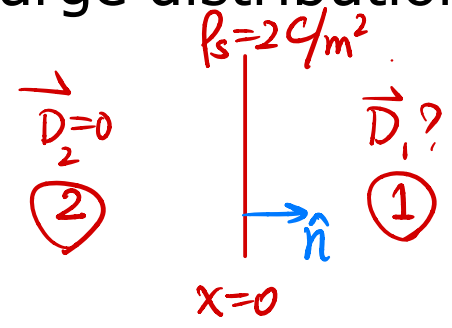


Example:
Free charges on the surface of a conductor

Finding D field

- Given $\vec{\boldsymbol{D}} = \mathbf{0}$ in the region $x < 0$ and $x > 5\,m$. Also, $x = 0$ and $x = 5$ m plane contain surface charge densities of $\rho_s = 2\,C/m^2$ and ρ_{so}. Determine ρ_{so} and $\vec{\boldsymbol{D}}$ for $0 < x < 5\,m$ if there are no other charge distributions.



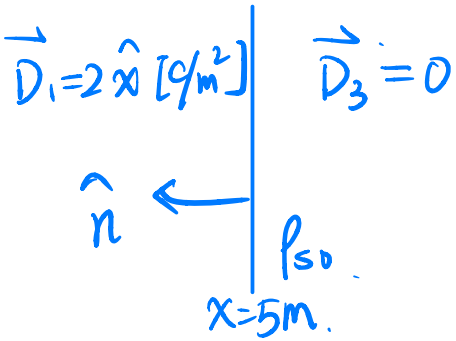


$$\hat{n} \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s$$

$$\hat{x} \cdot (\vec{D}_1 - 0) = \rho_s$$

$$D_{1x} = \rho_s = 2\,[C/m^2]$$

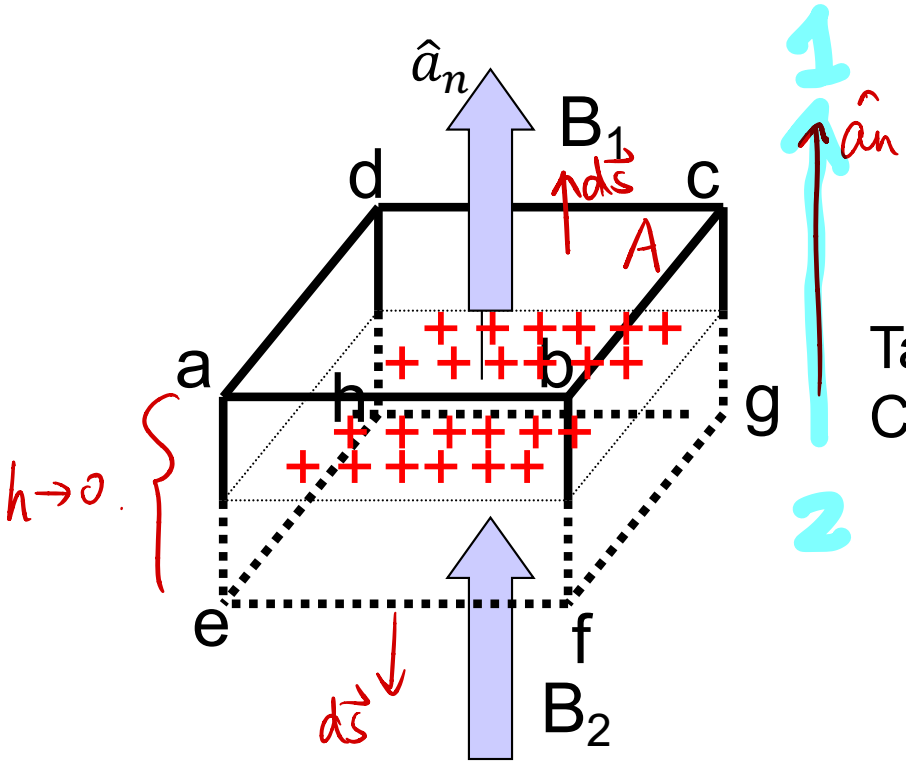
$$\vec{D}_1 = 2\hat{x}\,[C/m^2]$$



$$(-\hat{x}) \cdot (\vec{D}_1 - 0) = \rho_{so}$$

$$\rho_{so} = -\hat{x} \cdot 2\hat{x} = -2\,[C/m^2]$$

Gauss' Law for B at Boundary



$$\oiint_S \vec{B} \bullet d\vec{S} = 0$$

Top Bottom

$B_1 \cdot A - B_2 \cdot A = 0$

Take limit as *ae*, *bf*, *cg*, and *dh* go to zero

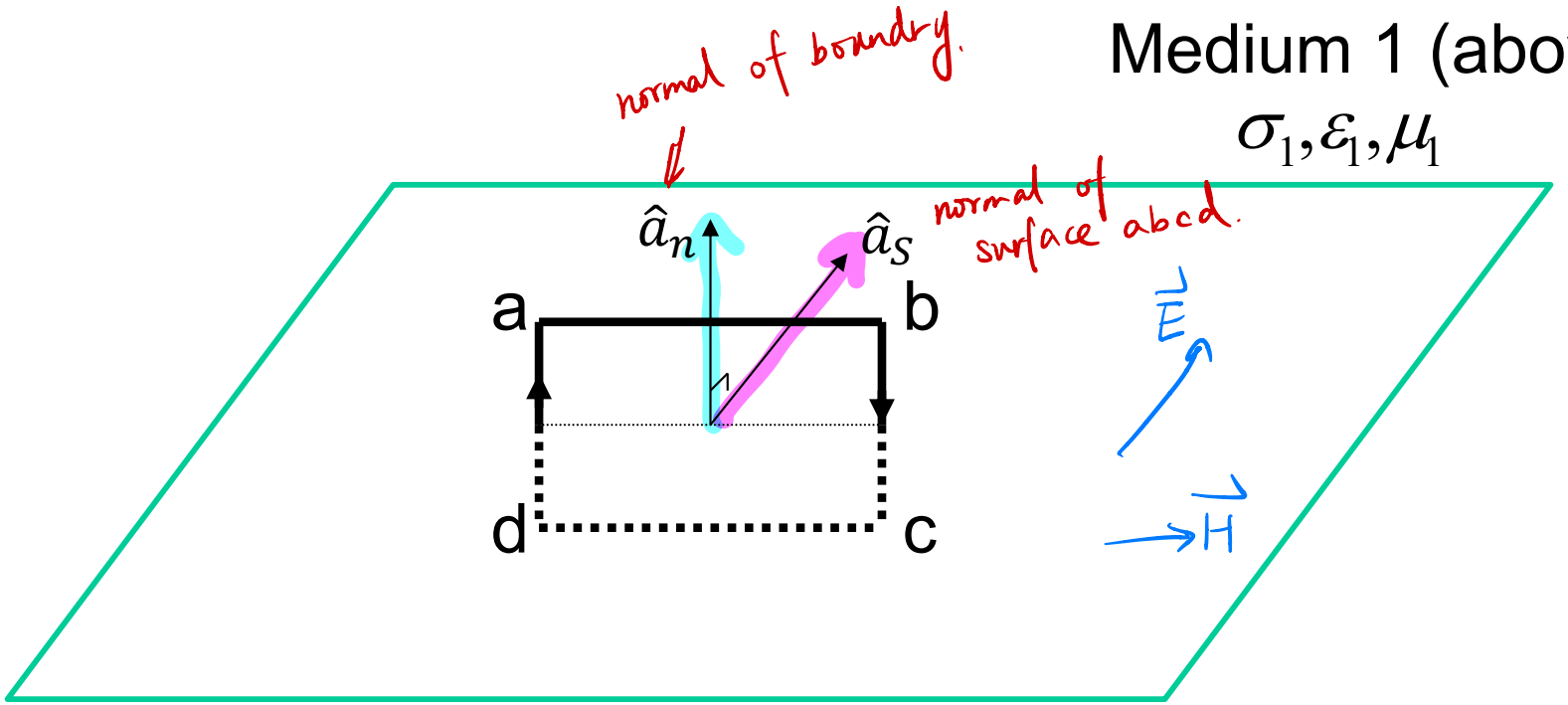
Consider B_1 and B_2 NORMAL TO SURFACE

$$B_{1n} = B_{2n} \text{ i.e. } B_n \text{ is continuous}$$

$$B_{n1} - B_{n2} = 0$$

$$\hat{a}_n \cdot \left(\vec{B}_1 - \vec{B}_2\right) = 0$$

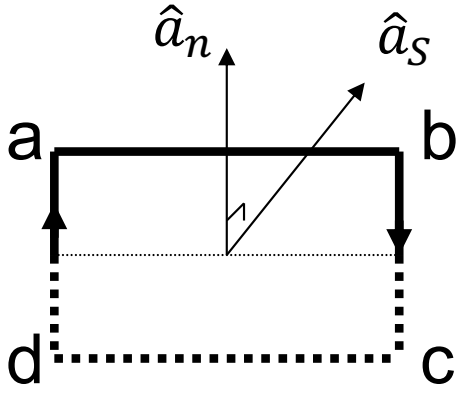
Closed Path Through a Boundary



Closed path: abcd

Apply Faraday's Law and Ampere's Law to the closed path

Normal Vectors



Medium 1 (above)
$\sigma_1, \varepsilon_1, \mu_1$

Medium 2 (below)
$\sigma_2, \varepsilon_2, \mu_2$

$\hat{\boldsymbol{a}}_n$ Vector NORMAL to the boundary. Points INTO medium 1

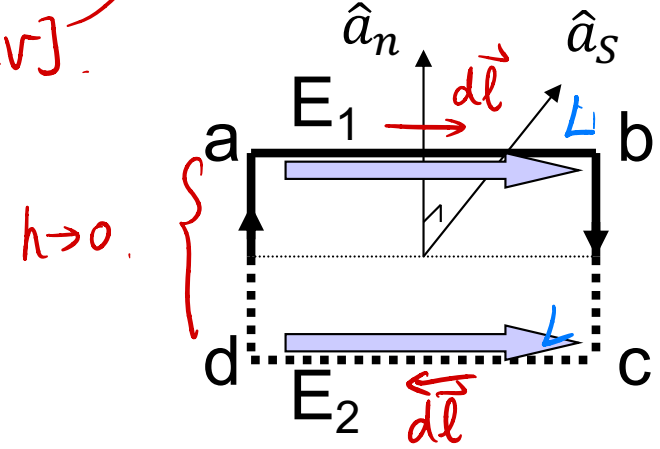
$\hat{\boldsymbol{a}}_S$ Vector normal to the path, TANGENT to the interface. (parallel)
Use right hand rule for path to define direction

Faraday's Law at Boundary

$$\oint_C \vec{E} \bullet d\vec{l} = -\frac{d}{dt}\iint_S \vec{B} \bullet d\vec{S} = 0$$

static $\frac{\partial}{\partial t} \to 0$

$[V]$



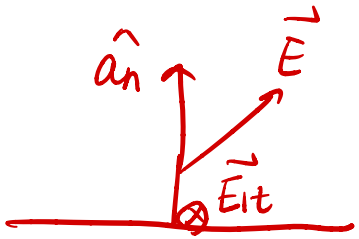
Top edge: $E_1 \cdot L$ — Bottom edge: $E_2 \cdot L$ $= 0$

Medium 1 (above)

$\sigma_1, \varepsilon_1, \mu_1$

Medium 2 (below)

$\sigma_2, \varepsilon_2, \mu_2$



Take limit as *ad* and *bc* go to zero

Consider remaining E_1 and E_2 TANGENT TO SURFACE

$$E_{1t} = E_{2t} \text{ i.e. } E_t \text{ is continuous}$$

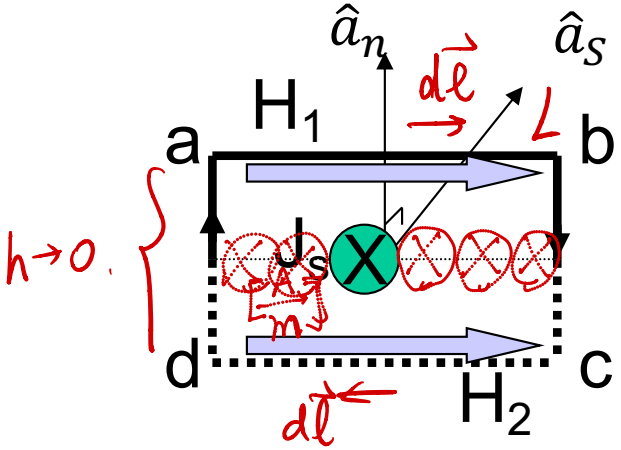
$$E_{t1} - E_{t2} = 0$$

$$\hat{a}_n \times \left(\vec{E}_1 - \vec{E}_2\right) = 0$$

Ampere's Law at Boundary

$$\oint_C \vec{H} \bullet d\vec{l} = \iint_S \vec{J} \bullet d\vec{S} + \frac{d}{dt} \iint_S \vec{D} \bullet d\vec{S} = I_{enclosed}$$





Medium 1 (above)

$\sigma_1, \varepsilon_1, \mu_1$

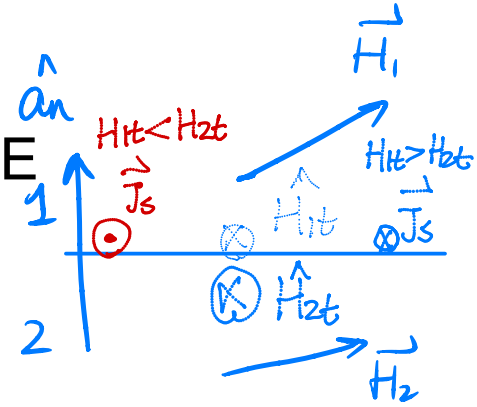
Medium 2 (below)

$\sigma_2, \varepsilon_2, \mu_2$

Take limit as *ad* and *bc* go to zero

Consider remaining H_1 and H_2 TANGENT TO SURFACE

$$H_{1t} - H_{2t} = J_S \text{ i.e. } H_t \text{ is discontinuous because of } J_S$$



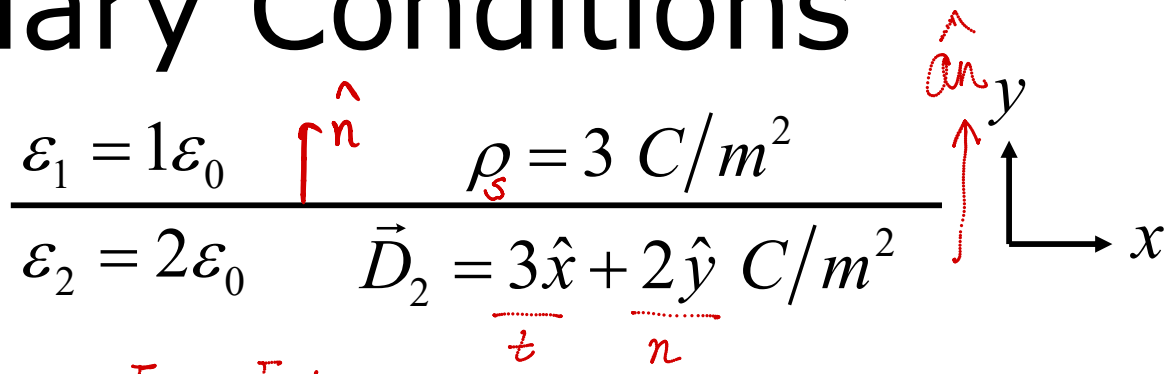
$$\vec{H}_{t1} - \vec{H}_{t2} = \vec{J}_s$$

$$\hat{a}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s$$

Note: we know nothing about H_n

Challenge Question: Boundary Conditions

- What is **D_1**?



$E_{1t} = E_{2t}$.

$\vec{D} = \varepsilon\vec{E} \rightarrow \vec{D}_t = \varepsilon\vec{E}_t$

$E_{1t} = E_{2t}$

$\frac{D_{1t}}{\varepsilon_0} = \frac{D_{2t}}{2\varepsilon_0} = \frac{3}{2\varepsilon_0}$.

$D_{1t} = \frac{3}{2}\hat{x}\ C/m^2$.

(a) $3\hat{x} + 2\hat{y}\ C/m^2$

(b) $3\hat{x} + 5\hat{y}\ C/m^2$

(c) $3\hat{x} - 1\hat{y}\ C/m^2$

(d) $\frac{3}{2}\hat{x} + 5\hat{y}\ C/m^2$

(e) $\frac{3}{2}\hat{x} - 1\hat{y}\ C/m^2$

$\hat{n}\cdot(\vec{D}_1 - \vec{D}_2) = \rho_s$.

$D_{1n} - D_{2n} = \rho_s$.

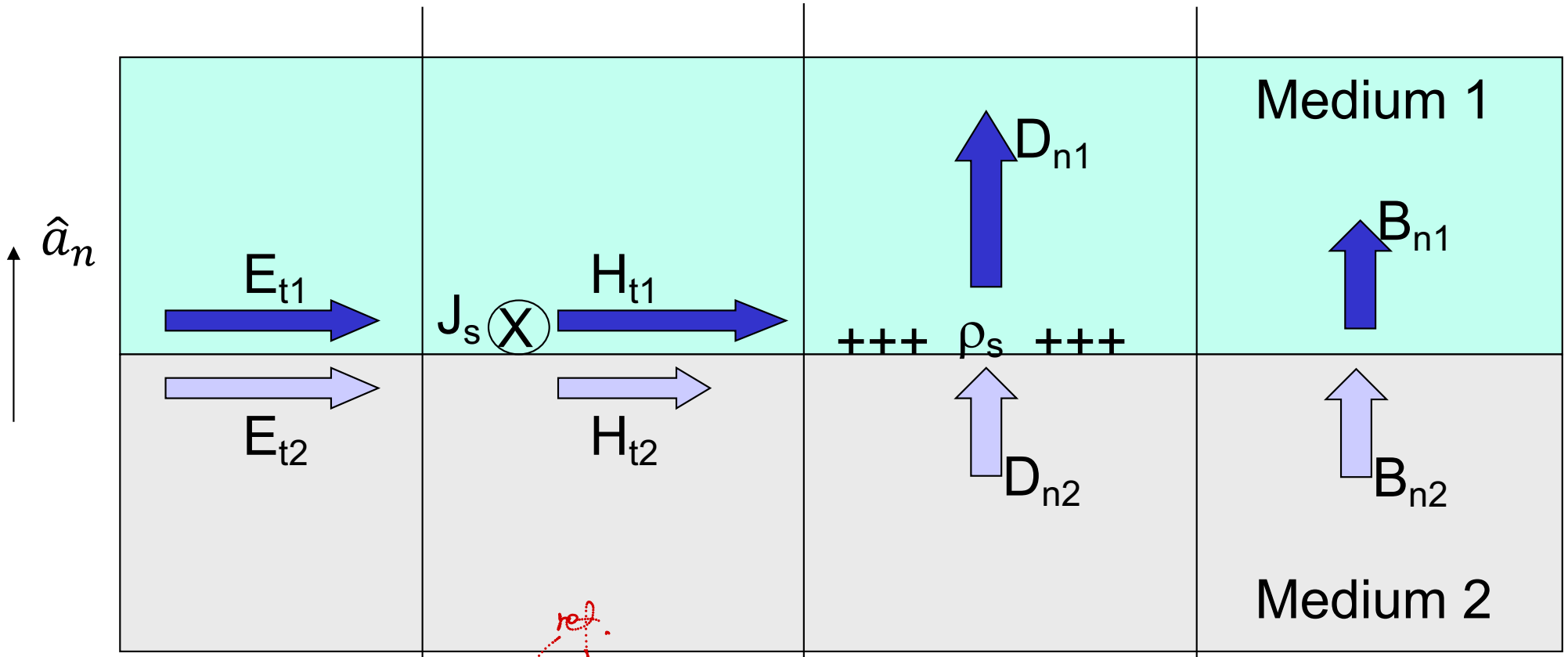
unknown

$D_{1n} - 2 = 3$

$D_{2n} = 5\hat{y}\ C/m^2$

Combine $\vec{D}_1 = \frac{3}{2}\hat{x} + 5\hat{y}\ C/m^2$.

Remember this drawing!!



$E_{t1} - E_{t2} = 0$ | $\vec{H}_{t1} - \vec{H}_{t2} = \vec{J}_s$ | $D_{n1} - D_{n2} = \rho_s$ | $B_{n1} - B_{n2} = 0$

$\hat{a}_n \times (\vec{E}_1 - \vec{E}_2) = 0$ | $\hat{a}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s$ | $\hat{a}_n \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s$ | $\hat{a}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0$

$\vec{D} = \varepsilon \vec{E}$ $\quad$ $\vec{B} = \mu \vec{H}$

How to memorize BC?

$\nabla \to \hat{n}$ $\quad D, E, B, H \to \vec{D}_1 - \vec{D}_2$

Differential form of Maxwell's Equations		Boundary Conditions
$\nabla \cdot \vec{\boldsymbol{D}} = \rho \to \rho_s$	$\longrightarrow$	$\hat{a}_n \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s$
$\nabla \times \vec{\boldsymbol{E}} = -\dfrac{\partial \vec{\boldsymbol{B}}}{\partial t}$	$\longrightarrow$	$\hat{a}_n \times (\vec{E}_1 - \vec{E}_2) = 0$
$\nabla \cdot \vec{\boldsymbol{B}} = 0$	$\longrightarrow$	$\hat{a}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0$
$\nabla \times \vec{\boldsymbol{H}} = \vec{\boldsymbol{J}} + \dfrac{\partial \vec{\boldsymbol{D}}}{\partial t}$	$\longrightarrow$	$\hat{a}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s$

J_s, $\quad \frac{\partial}{\partial t} \to 0$.

Laplacian Operator

$$\nabla \bullet \left(\nabla \Phi \right) = \nabla^2 \Phi$$

The "Laplacian" of a scalar field. (also called "Del Squared")

$\Phi(x,y,z)$ Scalar Field

$$\nabla \Phi = \frac{\partial \Phi}{\partial x} \hat{a}_x + \frac{\partial \Phi}{\partial y} \hat{a}_y + \frac{\partial \Phi}{\partial z} \hat{a}_z$$ Vector

$$\nabla^2 \Phi = \frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2}$$ Scalar

Poisson Equation for Potentials (static fields, constant ε)

$$\nabla \bullet (\nabla \Phi) = -\frac{\rho}{\varepsilon}$$

$$\nabla^2 \Phi = -\frac{\rho}{\varepsilon}$$

Known as "Poisson's Equation"

It's just Gauss' Law in terms of Potentials for a Static Field

If ρ=0, it is known as "Laplace's Equation"

Derivation of Poisson's & Laplace's equations

$$\nabla \bullet \mathbf{D} = \rho, \quad \mathbf{D} = \varepsilon \mathbf{E}$$

$$\mathbf{E} = -\nabla V$$

$$\nabla \bullet (\varepsilon \mathbf{E}) = -\nabla \bullet (\varepsilon \nabla V) = \rho$$

for homogeneous medium

$$\varepsilon \nabla \bullet \nabla V = \varepsilon \nabla^2 V = -\rho$$

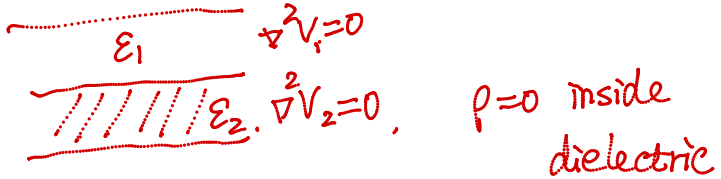
$$\nabla^2 V = -\frac{\rho}{\varepsilon} \quad \text{(Poisson's equation)}$$

$$\nabla^2 V = 0 \quad \text{(Laplace's equation)}$$

$$\nabla \bullet \nabla V = \nabla \bullet \left(\hat{\mathbf{x}} \frac{\partial V}{\partial x} + \hat{\mathbf{y}} \frac{\partial V}{\partial y} + \hat{\mathbf{z}} \frac{\partial V}{\partial z} \right)$$

$$= \left(\hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z} \right) \bullet \left(\hat{\mathbf{x}} \frac{\partial V}{\partial x} + \hat{\mathbf{y}} \frac{\partial V}{\partial y} + \hat{\mathbf{z}} \frac{\partial V}{\partial z} \right)$$

$$= \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

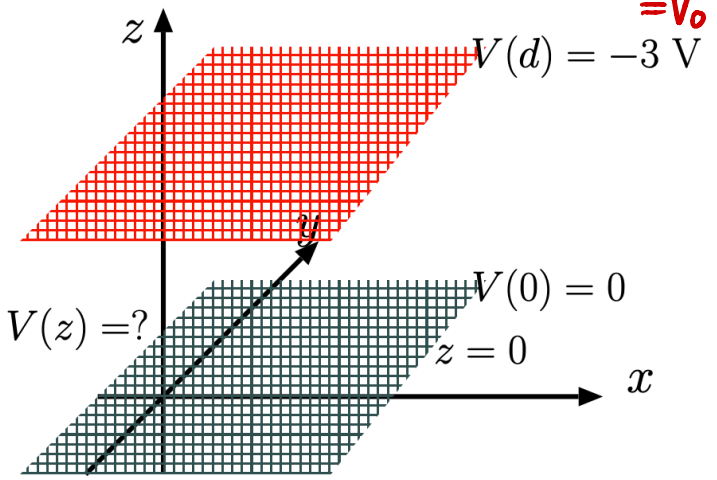


Laplace operator has different forms for different coordinate systems.

Example

Consider a pair of parallel conducting metallic plates separated in z direction by distance of d = 2 m.

There is electrostatic field $\vec{E} = E_z \hat{z}$ measured in between the plates. Using Laplace's equation determine V (z) and E(z) between the plates if the potential of the plate at z = 0 is 0 (the ground), while the potential of the plate at z = d is −3 V.



$\propto z<d$ air → no charge.

$\nabla^2 V = 0$

$\frac{\partial^2}{\partial x^2}V + \frac{\partial^2}{\partial y^2}V + \frac{\partial^2 V}{\partial z^2} = 0$

$\frac{\partial V}{\partial z} = A$. $\quad V(z) = Az + B$

use. $V(z=0)=0 \rightarrow V(0) = B = 0$

$V(z=d)=V_0 \rightarrow V(d) = Ad = V_0 \Rightarrow A = \frac{V_0}{d}$

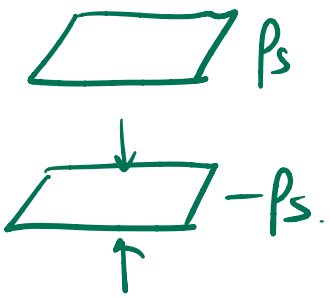
$$V(z) = \frac{V_0}{d} z \quad [V]$$

$\vec{E} = -\nabla V = -\left(\frac{\partial}{\partial x}V, \frac{\partial V}{\partial y}, \frac{\partial V}{\partial z}\right) = -\left(0, 0, \frac{V_0}{d}\right)$

$$\vec{E} = -\frac{V_0}{d}\hat{z} \quad [V/m]$$

Example (cont.)

what are the surface charge densities of the metallic plates located at z = 0 and z = 2 m surfaces?



Top $\vec{D}_1 = \frac{\rho_s}{2}\hat{z}$ $z>d$.

$\vec{D}_2 = -\frac{\rho_s}{2}\hat{z}$

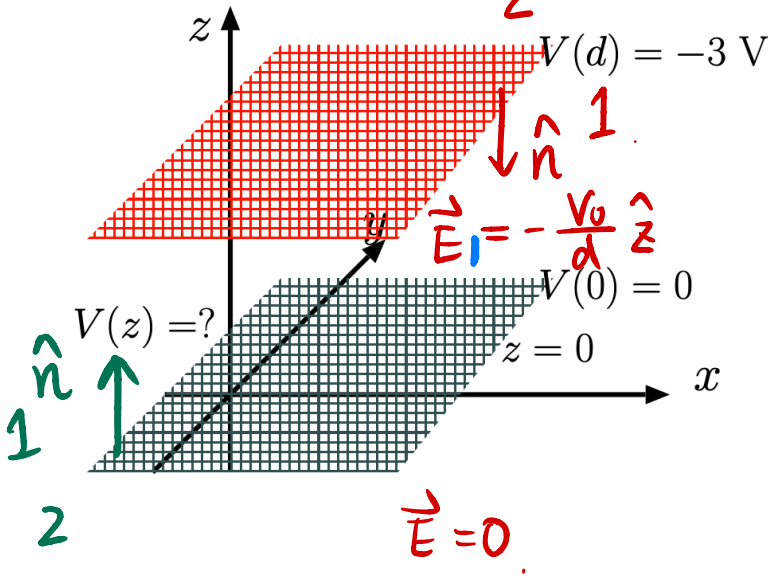
$\vec{D} = \vec{D}_1 + \vec{D}_2 = 0$.

$\vec{E} \rightarrow \vec{D} \rightarrow \rho_s$.

Top plate. $\hat{n}\cdot(\vec{D}_1 - \vec{D}_2) = \rho_s$ $\vec{D} = \varepsilon_0\vec{E}$.

$-\hat{z}\cdot\left(\varepsilon_0\cdot(-\frac{V_0}{d}\hat{z}) - 0\right) = \rho_s$.

$\rho_s = \varepsilon_0\frac{V_0}{d}$ [C/m²].

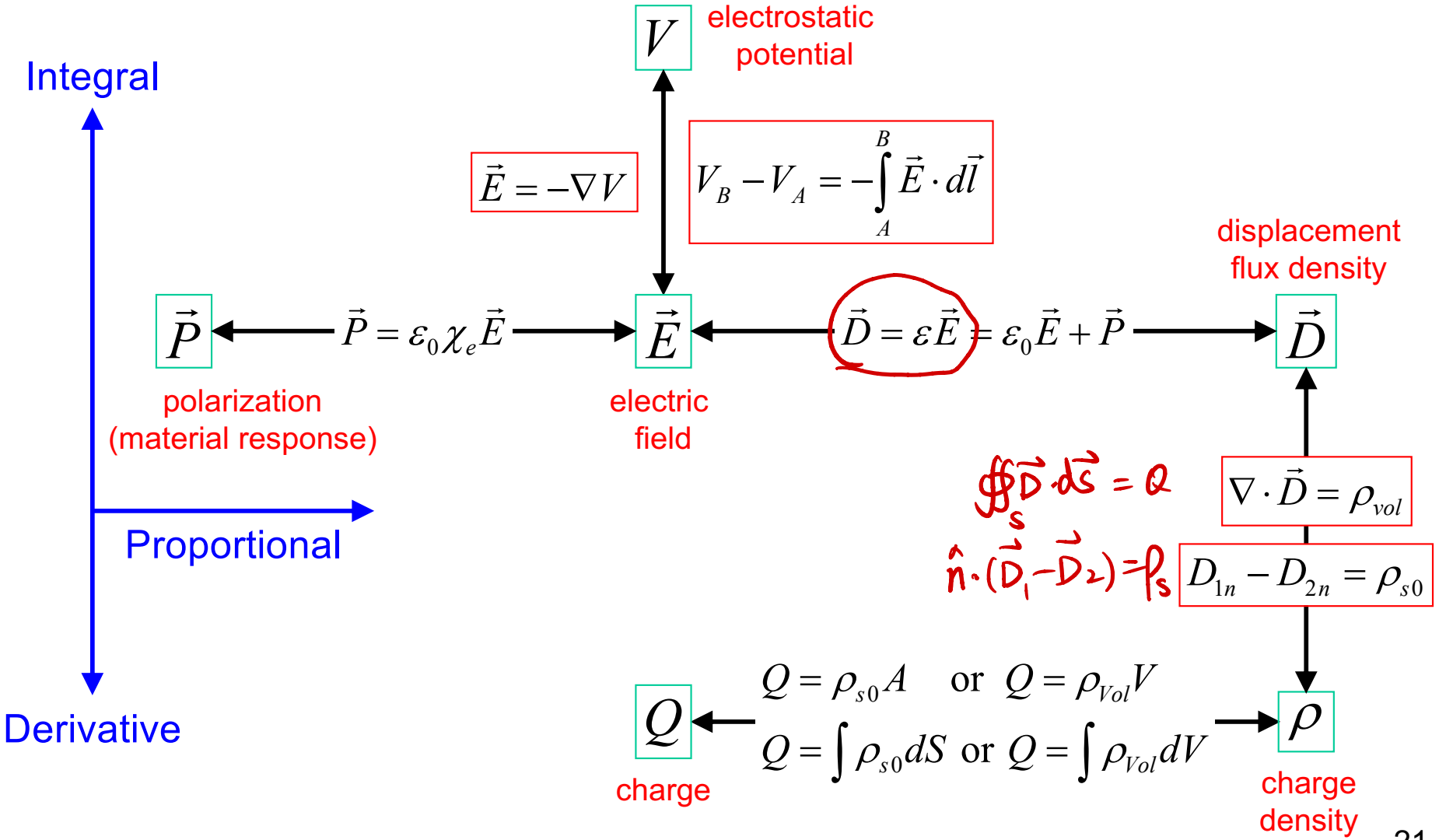


Bottom plate. $\hat{n}\cdot(\vec{D}_1 - \vec{D}_2) = \rho_s'$

$\hat{z}\cdot\left(\varepsilon_0\cdot(-\frac{V_0}{d}\hat{z}) - 0\right) = \rho_s'$

$\rho_s' = -\varepsilon_0\frac{V_0}{d}$ [C/m²].

Connection of Concepts for Electrostatics



Summary

- Boundary conditions

$$E_{t1} - E_{t2} = 0 \qquad \vec{H}_{t1} - \vec{H}_{t2} = \vec{J}_s \qquad D_{n1} - D_{n2} = \rho_s \qquad B_{n1} - B_{n2} = 0$$

$$\hat{a}_n \times (\vec{E}_1 - \vec{E}_2) = 0 \quad \hat{a}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s \quad \hat{a}_n \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s \quad \hat{a}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0$$

- For static fields, if ε is constant, Φ satisfies Poisson's equation:

$$\nabla^2 \Phi = -\frac{\rho}{\varepsilon}$$